

Practice Quiz 06C — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 06

Find a missing hypothesis and repair an existence-and-uniqueness proof.

1. A student claims that for every real a there is exactly one real x with $ax = a$, and gives this argument:

Divide both sides by a to obtain $x = 1$. Therefore there is exactly one solution.

- (a) Identify the invalid step, and describe all solutions when $a = 0$.

Solution The division by a is invalid when $a = 0$. In that case the equation is $0x = 0$, which every real x satisfies. Thus existence holds but uniqueness fails.

- (b) Add a hypothesis on a that makes the claim true, and prove the corrected claim. Label the existence and uniqueness parts.

Solution The corrected claim is: for every real $a \neq 0$, there is exactly one real x such that $ax = a$. Let such an a be arbitrary.

Existence. Set $x = 1$. Then $ax = a \cdot 1 = a$.

Uniqueness. If real numbers x and y both satisfy the equation, then $ax = a = ay$. Since $a \neq 0$, division by a gives $x = y$. Thus the claim holds for every nonzero real a . ■